

$$1) \frac{dy}{dx} = 3x - 1$$

$$dy = (3x - 1) dx$$

$$dy = 3x dx - dx$$

$$y = \frac{3x^2}{2} - x + C$$

$$2) y dx - x dy = 0$$

$$\frac{dx}{x} = \frac{dy}{y}$$

$$\ln k + \ln x = \ln y$$

$$y = kx$$

$$3) \frac{dy}{dx} + y \cos x = 0$$

$$\frac{dy}{dx} = -y \cos x$$

$$\frac{dy}{y} = -\cos x dx$$

$$\ln y = -\sin x + \ln k$$

$$y = e^{k(-\sin x)}$$

$$4) (x - 1) dy - y dx = 0$$

$$(x - 1) dy = y dx$$

$$\frac{dy}{y} = \frac{dx}{x - 1}$$

$$\ln y = \ln |x - 1| + \ln k$$

$$y = k(x - 1)$$

$$5) (4y + yx^2) dy - (2x + xy^2) dx = 0$$

$$y(4 + x^2) dy = x(2 + y^2) dx$$

$$\frac{y dy}{(2 + y^2)} = \frac{x dx}{(4 + x^2)}$$

$$\ln |2 + y^2| = \ln |4 + x^2| + \ln C$$

$$2 + y^2 = C(4 + x^2)$$

$$y^2 = C(4 + x^2) - 2$$

$$6) \sin x (e^{-y} + 1) dx = (1 + \cos x) dy; x = 0, y = 0$$

$$\frac{\sin x}{1 + \cos x} dx = \frac{dy}{e^y + 1}$$

$$\ln |C| - \ln |1 + \cos x| = \ln |1 + e^y|$$

$$C = (1 + \cos x)(1 + e^y)$$

$$t = 2 + y^2 \quad dt = 2y dy$$

$$\int \frac{y dt}{t 2y}$$

$$\frac{dy}{\frac{1}{e^y} + 1} = \frac{dy}{\frac{1 + e^y}{e^y}}$$